

RAJESH VHANKADE

AI/ML Engineer | Generative AI | Agentic AI | RAG | LangChain | LangGraph | MCP | FastAPI | Python | AWS

+91 9763263822 | vhanakadenrajesh@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Pune, Maharashtra, India

PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years of experience designing and deploying Generative AI, Retrieval-Augmented Generation (RAG), and Agentic AI solutions using LLMs, LangChain, LangGraph, MCP, FastAPI, and AWS. Experienced in building enterprise-grade document intelligence systems, semantic search platforms, AI-powered assistants, and workflow automation solutions. Skilled in developing scalable backend services, vector search architectures, tool-calling workflows, and production-ready AI applications that improve information access and operational efficiency.

TECHNICAL SKILLS

- **Generative AI & LLMs:** OpenAI GPT-4, Anthropic Claude, Google Gemini, Llama Models, Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph, MCP (Model Context Protocol), Prompt Engineering, Tool Calling, Function Calling, AI Agents, Structured Output Generation, Hugging Face Transformers
- **AI/ML & NLP:** Natural Language Processing (NLP), Embeddings, Semantic Search, Similarity Search, Information Retrieval, Text Classification, Vector Search
- **Retrieval & Vector Databases:** FAISS, ChromaDB, Hybrid Search, Metadata Filtering, Re-ranking, Vector Indexing
- **Backend Development & APIs:** Python, FastAPI, REST APIs, Asynchronous Programming, API Integration
- **Databases:** PostgreSQL, MongoDB, SQLite
- **Cloud & DevOps:** AWS, Docker, CI/CD, Application Deployment
- **Tools & Platforms:** Git, GitHub, Postman, VS Code, Jupyter Notebook

PROFESSIONAL EXPERIENCE

SmartMatrix Digital Services Pvt Ltd. | April 2024 - Present

AI/ML Engineer

- Designed and deployed Generative AI applications for enterprise knowledge management and document intelligence use cases.
- Built scalable FastAPI-based backend services supporting AI-powered search, question-answering, and workflow automation solutions.
- Developed Retrieval-Augmented Generation (RAG) pipelines covering document ingestion, chunking, embedding generation, vector indexing, retrieval, and response generation.
- Implemented semantic search, hybrid retrieval, metadata filtering, and re-ranking techniques, improving retrieval relevance by approximately 25%.
- Integrated OpenAI and Hugging Face models into production workflows using prompt engineering and tool-calling mechanisms.
- Designed LangGraph-based workflow orchestration for multi-step retrieval and response generation processes.
- Built MCP-enabled integrations to connect AI systems with external tools and business workflows.
- Optimized application performance using asynchronous processing, reducing response latency by approximately 40%.
- Containerized and deployed AI services using Docker and AWS infrastructure.
- Collaborated with cross-functional teams to translate business requirements into deployable AI solutions.

Radixile Technology Solutions | Oct 2023 – Mar 2024

Machine Learning Engineer Intern

- Assisted in end-to-end machine learning development including data preprocessing, feature engineering, model training, validation, and performance optimization.

- Developed and evaluated predictive models using Python-based ML libraries and statistical techniques.
- Supported integration of ML functionality into business applications and internal products.
- Participated in experimentation and model optimization initiatives to improve prediction accuracy and reliability.

PROJECT EXPERIENCE

Enterprise Knowledge Assistant

Domain: Manufacturing Operations & Knowledge Management

- Developed an AI-powered enterprise assistant enabling teams to search and access information from SOPs, maintenance manuals, troubleshooting guides, and operational documentation.
- Built a complete RAG architecture supporting ingestion, chunking, embedding generation, vector indexing, retrieval, and answer generation.
- Implemented semantic search and metadata-aware retrieval to improve information discovery across large document repositories.
- Designed prompt templates and grounding strategies to reduce hallucinations and improve response accuracy.
- Developed FastAPI services exposing document search and question-answering APIs.
- Improved response performance through asynchronous processing and caching mechanisms.
- Deployed the solution using Docker and AWS EC2.
- Tech Stack: Python, FastAPI, OpenAI API, LangChain, FAISS, PostgreSQL, Docker, AWS EC2

HR-Payroll Data Synchronization Platform

Domain: HR & Payroll Process Automation

- Developed a prototype system to synchronize employee data changes between HR and Payroll databases using MCP (Model Context Protocol).
- Created MCP servers exposing tools for detecting employee record changes, validating updates, generating sync payloads, and applying changes to the target system.
- Implemented SQLite triggers to automatically capture employee updates and store change records for processing.
- Built workflow logic to validate employee changes before transferring data between systems.
- Generated structured JSON payloads containing employee updates and synchronization details.
- Applied approved updates to the payroll database to keep employee information consistent across systems.
- Simulated a common business scenario where salary and employee profile updates made in HR systems are reflected in Payroll systems.
- Tech Stack: Python, LangChain, MCP (Model Context Protocol), SQLite, JSON, Git

PROFESSIONAL DEVELOPMENT

Data Science & Machine Learning Program | Yess Infotech, Pune | Sep 2022 - Sep 2023

- Completed training in Python, Statistics, Machine Learning, Deep Learning, NLP, and Data Science.
- Built projects involving data preprocessing, model training, classification algorithms, and evaluation techniques.
- Gained practical experience with machine learning workflows and deployment concepts.

EDUCATION

- **Bachelor of Engineering (Computer Engineering)**
Savitribai Phule Pune University | 2022
- **Diploma in Computer Engineering**
Government Polytechnic, Miraj | 2019